Subject Dependent Feature Extraction Method for Motor Imagery based BCI using Multivariate Empirical Mode Decomposition

Jin Zhang, Chungang Yan, Xiaoliang Gong
College of Electronic and Information Engineering, Tongji University, Shanghai 201804, China
Key Laboratory of Embedded System and Service Computing (Tongji University), Ministry of Education, Shanghai, China
Shanghai Electronic Transactions and Information Service Collaborative Innovation Center

Abstract— Extracting efficient features is an important stage in Brain Computer Interface (BCI). In motor imagery (MI) based BCI, features that characterize ERS/ERD phenomenon could favor the discriminant of different kinds of MI tasks. Common Spatial Pattern (CSP), a spatial filtering method which is often used in MI-EEG feature extraction, shows outstanding performance due to its effectiveness in maximizing the difference of variance between each class. But one of the shortcomings of CSP is that performance is greatly affected by the frequency of the signal, because the variances can be used to discriminate each class only by processing those task related frequency bands. However, individual differences existed in EEG is an obstacle for CSP. Although MI related frequency bands focus on mu and beta rhythms, for every subject, the distinguishing bands could be different. In this paper, a novel feature extraction method (Subject Dependent Multivariate Empirical Mode Decomposition, SD-MEMD) for MI based BCI is proposed, it utilizes MEMD algorithm to decompose the multi-channel EEG into a set of Intrinsic Mode Functions (IMFs), each IMF represents an inherent oscillation mode of the raw signal. Then an enhanced EEG is re-constructed after a careful selection of the task related IMF subset. Classification accuracy for right hand and left hand MI tasks is improved by 5.76% on our dataset.

Key Words—Brain Computer Interface, motor imagery, Common Spatial Pattern, Multivariate Empirical Mode Decomposition

I. Introduction

Brain Computer Interface (BCI) can help people control outer equipment by translating physiological signals into certain commands, so it provides another choice for communication between human and the outer equipment. Quite a lot of BCI studies use electroencephalograph (EEG) as input signal in the safety and good spatial temporal resolution considerations. However, some characteristics, like low signal-to-noise-ratio(SNR), non-stationary, big individual differences, contribute to the complexity of EEG. Thus, extracting discriminant features of EEG signals is a challenging task.

Motor imagery (MI) is commonly used as one of the experiment paradigms in EEG-based BCI [1,2]. Subjects can generate different kinds of EEG by executing infinite movement imagery tasks. According to knowledge of neuroscience, when people are doing MI tasks, there will be activation around the contralateral area and simultaneously suppression will occur around the ipsilateral area, so called Event-Related Synchronization (ERS) and Event-Related De-synchronization (ERD). And this phenomenon mainly appears in the frequency band of mu (8-13Hz) and beta (14-30Hz) in the EEG signal. What`s more, MI-based BCI usually arrange many electrodes around the sensorimotor cortex over the scalp in order to collect those task related signals as thorough as possible. So the experimental signals consist of many dimensions. Translating MI EEG relies on diagnosing ERS/ERD embedded in the signals from multiple channels, therefore, the more efficient features extracted to characterize ERS/ERD, the more accurate the recognition of EEG.

Among many feature extraction algorithms, Common Spatial Pattern (CSP) [3] has proved outstanding performance in MI-based BCI. Generally, CSP obtains several spatial patterns by solving an eigenvalue decomposition problem. When signal projects on certain spatial patterns, differences of variance between each class will be maximized. However, one of the disadvantages of original CSP is that the performance relies heavily on the range of frequency band embedded in EEG, because the variances can be used to discriminate each class only by processing those task related frequency bands [4]. In the mean time, one trait that needs to be noticed is that for every subject, energy distributes unequally in each narrowband due to the individual differences. Although MI related frequency bands focus on mu and beta rhythms, for every subject, the distinguishing bands could be different. In [4], Filter Bank Common Spatial Pattern (FSCSP) has been proposed to narrow the frequency band of the signal, aiming at enhancing the performance of CSP. It divides the raw EEG signals into a set of sub-signals containing a sub frequency band, then extract features from each sub-signals respectively. However, this method ignored the individual differences, so observant selection of frequency narrowband for each person is necessary.

Empirical Mode Decomposition (EMD) [5] algorithm is specialized at processing non-linear and non-stationary signals like EEG. It decomposes temporal signals into infinite zero-mean components (i.e. Intrinsic Mode Functions, IMF), each component is modulated by sub-band frequency and amplitude information, so it can reflect inherent oscillation mode of the original signal. Many studies have combined EMD and CSP to analyze MI-based EEG [6,7], but they ignored the natural correlation between channels. Actually, when it comes to multivariate problems, EMD is not appropriate because unique numbers or properties (frequency) of IMFs might show up when processing different variables. So it`s complicated to choose IMFs from these diverse results.

To address this difficulty, in 2009, N.Rehman and D.P.Mandic published an algorithm named Multivariate Empirical Mode (MEMD) [8]. And in [9,10], authors have

978-1-5386-3142-3/17/$31.00 ©2017 IEEE

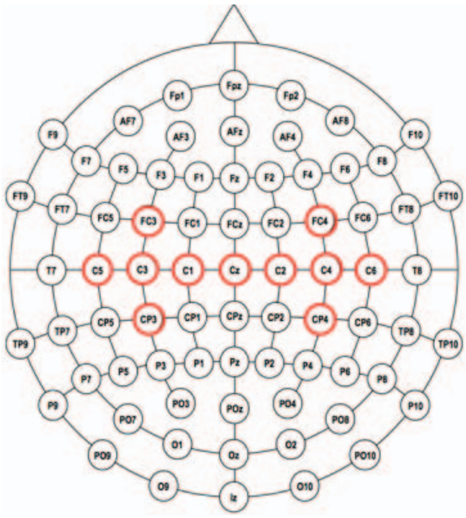

Fig. 1. Electrodes locations for this experiment and locations with red circles stand for selected

applied this method to process multi-channel EEG and proved that MEMD could get the same number of IMFs of similar frequencies from multiple channels. But their work didn't pay much attention to individual differences existed in EEG, therefore improvement could be achieved when this problem has been taken into account.

In this paper, we utilized MEMD to decompose multi-channel EEG at the first stage, discovered the alignment in IMF level in concern of their embedded amplitude and frequency properties, which further proved the efficiency of MEMD in processing multivariate EEG, and was in consistent of Cheolsoo`s results. At the second stage, an effective subject dependent (SD) feature extraction method was proposed for a two classes (i.e. right hand and left hand) motor imagery problem. IMFs which reflected ERS/ERD phenomenon and were within the mu and beta band scope were selected to composite an enhanced EEG. So the new signal was filtered out of artefacts and task unrelated components. Then Support Vector Machine (SVM) was used to classify different EEG according to CSP features. Ten-fold cross validation was applied during classification.

The structure of this paper is as follows. Section II describes the procedures and specification for data collection. Section III reviews the MEMD algorithm and its necessity in processing multi-channel EEG. Section IV explains the SD-MEMD method and section V describes the feature extraction and classification stages. Finally, section VI concludes this paper.

II. Data Collection

There were totally 14 subjects participated in our MI based tasks, aging varying 23.7 ± 2.3 and they were all right handed. ActiCHamp 64 channels was used as the bio-signal amplifier to collect scalp EEG. Experimental channel set consists of 11 channels according to the standard International 10-20 system, their locations were in Fig.1. In the Recorder software setup, sample rate was set to be 256Hz, and the recorded data was filtered in advance with a band pass filter of 0.5-40Hz. All the subjects were asked to sit in a comfortable chair in front of a screen.

During the experiment, an individual should complete 3 sessions in total, each session consisted of 20 trials, a short break was arranged between two sessions. At the start of each session, there was a fixation cross on the screen, lasting for 2 seconds. Thereafter, twenty cues showed up in succession, they were either right arrows or left arrows, and subjects should imagine movement of their right or left hands according to the arrows. Each trial lasted for 2 seconds.

III. Multivariate Empirical Mode Decomposition

Since EEG has some traits like low SNR, non-stationary and large individual differences, and only mu and beta bands are related to MI tasks, so using MEMD to filter the signal can improve the quality and make preparation for further analysis.

MEMD is developed from EMD method to handle multivariate signals simultaneously. It is a natural extension of EMD. EMD algorithm decomposes the original signal $x(t)$ into a limited number of IMFs $C_i(t)$and a residue $r(t)$, expressed as follows:

$$x(t) = \sum_{i=1}^{M} C_i(t) + r(t) \quad (1)$$

where t and M stand for the time points and the number of IMFs respectively. In order to get the IMFs, EMD finds all local maxima and local minima of the signal, and computes the envelope by interpolating those local extremes. Then it removes the mean of the envelope from the signal, and repeats these steps until the remaining component meets two conditions: (1) the number of extremes and the number of zero crossings differ at most by one, (2) the mean of the envelopes is approximately zero. So each IMF can reflect one of the inherent oscillating mode, or intrinsic character of the original signal. However, for multi-channel EEG, using EMD can be difficult, because EMD only manipulates temporal EEG from a single channel in one time, so each IMF set obtained from each channel might be unique, no matter quantity or containing frequencies.

So MEMD has been proposed to address this problem. The major obstacle of MEMD is that the local extremes for all the channel's data are difficult to find, and the solution is to calculate the envelopes of multivariate signals by using real-valued projections along multiple directions on hyperspheres. Comparing to processing each channel using EMD successively, MEMD has fully considered the cross channel information, so it can avoid the unique problem, in the mean time, amplitude and frequencies properties from different channels will show alignment in IMF level.

Figure 2 can prove the efficiency of MEMD. A typical subject (Sub.3) was chosen for this part of analysis. Figure 2 (a) and (b) plot the first five IMFs from C3 and C4 channels when the subject was doing a right hand imagery task and a left hand imagery task. Y-axis stand for the amplitude of each IMF. As we can see in Fig.2 (a), for the same IMF of each channel, the inherent frequencies and amplitude are similar, so the alignment in IMF level could be proved. Under this alignment character,

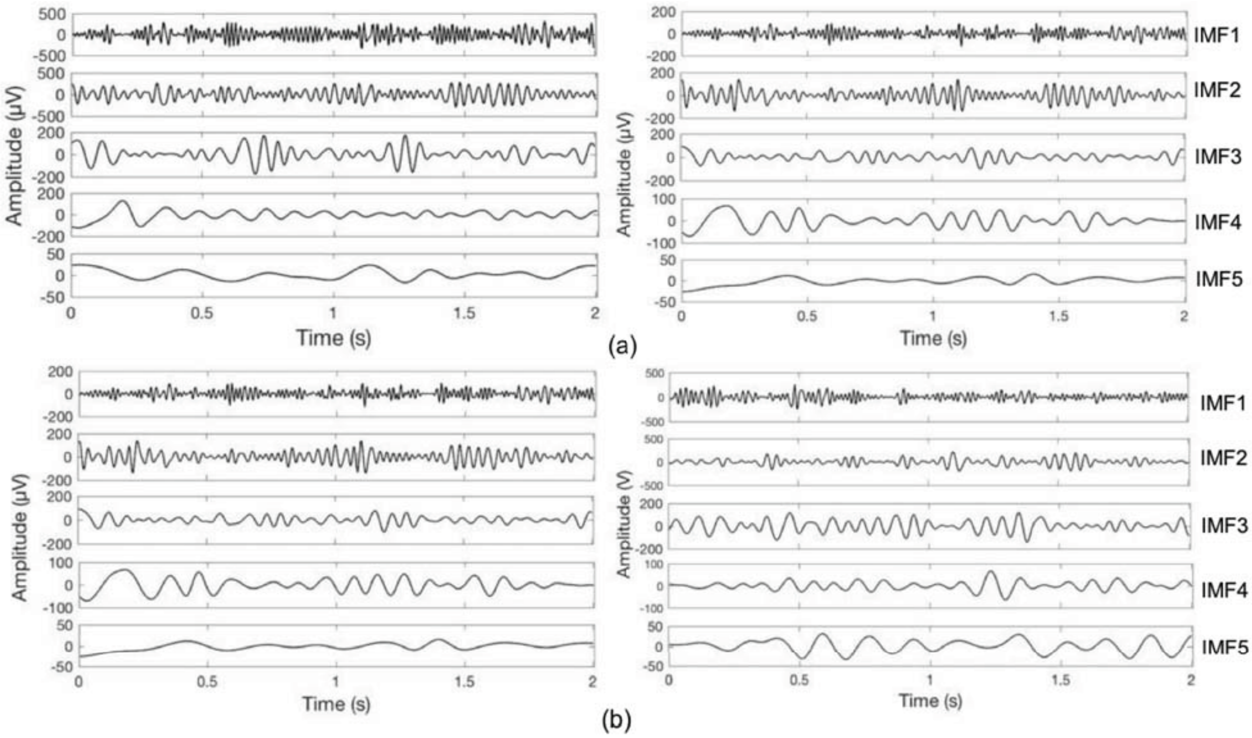

Fig. 2. The first 5 IMFs from C3 and C4 channels when Sub.3 was doing a right hand imagery task (a) and a left hand imagery task (b).

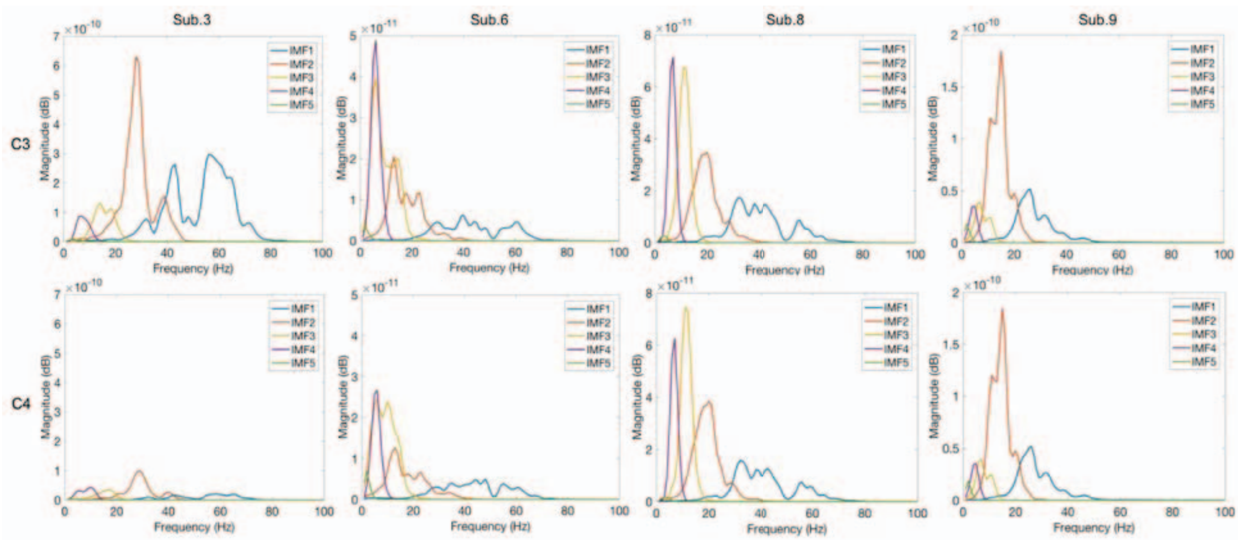

Fig. 3. Frequency and magnitude information of each IMF from C3 and C4 for Sub.3 Sub.6 Sub.8 and Sub.9 when they were doing a right hand imagery task.

choices for certain IMFs could be achieved for all the channels. And what`s more, since C3 located in the left hemisphere and C4 located in the right hemisphere, noticing that amplitude of C3 was much higher than that of C4, a clear ERS/ERD phenomenon could be observed. For Fig.2 (b), vice sera.

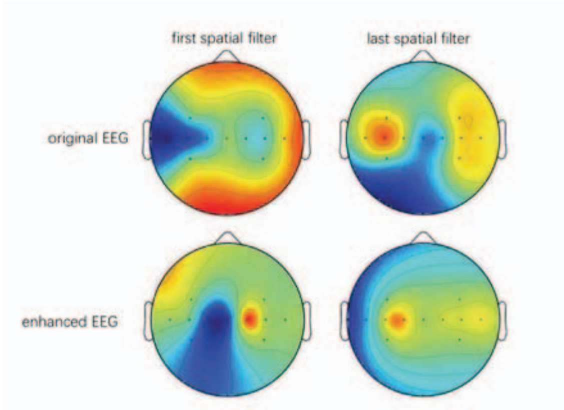

Fig. 4. A top view for the power distribution of the first and last spatial filter over the scalp from Sub.6

TABLE I. For each subject, column 2 represents the chosen IMF subset using SD-MEMD and column 3 and 4 represents the mean accuracy with standard derivation from original EEG and enhanced EEG

Subject	IMF subset	Original EEG	Enhanced EEG
Sub.1	{2,3,4}	63.29 ± 12.47	80.56 ± 11.19
Sub.2	{3,4}	67.22 ± 12.13	82.22 ± 8.20
Sub.3	{2,3,4}	100 ± 0	100 ± 0
Sub.4	{2,3,4}	86.11 ± 11.79	92.22 ± 11.79
Sub.5	{2,3,4}	100 ± 0	100 ± 0
Sub.6	{2,3,4}	57.78 ± 12.61	72.22 ± 8.69
Sub.7	{2,3}	69.41 ± 12.31	70.00 ± 13.66
Sub.8	{2,3,4}	65.00 ± 10.81	62.37 ± 11.30
Sub.9	{1,2,3}	71.33 ± 11.05	80.86 ± 12.27
Sub.10	{2,3,4}	100 ± 0	100 ± 0
Sub.11	{2,3,4}	72.22 ± 10.14	74.44 ± 12.60
Sub.12	{1,2}	93.33 ± 5.14	95.78 ± 3.72
Sub.13	{2,3}	61.11 ± 8.38	67.50 ± 7.97
Sub.14	{1,2}	90.83 ± 6.45	93.33 ± 3.40
$mean \pm std$		78.40 ± 15.84	$\mathbf{84.16 \pm 12.81}$

IV. Subject Dependent MEMD

In this section, we chose four out of fourteen typical subjects (i.e. Sub.3, Sub.6, Sub.8, Sub.9) to explain the Subject Dependent MEMD (SD-MEMD) method. Figure 3 showed the frequency and magnitude information of each IMF from C3 and C4 when the subjects were doing a right hand imagery task. It can be observed apparently that for the same motor imagery task, there was a big difference among different subjects in the frequencies and magnitude. For Sub.3, magnitude of IMF1-4 from C3 was much larger than that from C4, even if the frequency band of IMF1 was not in the scope of mu and beta. So no matter which IMFs we chose, the type of the task could be predicted easily. For Sub.6, only IMF1-3 showed less significant difference and were also within mu and beta band, so for this typical subject, only these IMFs could be used to re-construct the enhanced EEG. However, Sub.8 appeared an utterly different situation. No disparity showed up in any IMF, so no ERS/ERD could be observed. In could be inferred that this typical subject couldn`t perform a good control for this kind of motor imagery task, so called “BCI blind”. Therefore, classification for MI-EEG from this kind of people has a lot of randomness. For Sub.9, energy of the first IMF has been within beta band already, so the discriminant subset was IMF1-2. To conclude these observations above, individual differences existed in the EEG did have an impact on the IMF subset selection, and this character could further affect the signal quality. So for each subject, we first decomposed the multi-channel EEG into a set of IMFs using MEMD, and then the work of IMF subset selection was subject dependent, at last, enhanced EEG was obtained by simply adding the IMF subset together. Chosen IMF subset for each subject was shown in Table I.

V. Feature Extraction and Classification

After filtering by SD-MEMD method, EEG signal of each person was re-constructed. Comparing to the original EEG, new signals were removed by artefacts and unrelated component toward the motor imagery tasks. Then Common Spatial Pattern (CSP) and Support Vector Machine (SVM) were applied in the feature extraction and classification stages.

Common Spatial Pattern (CSP) is a spatial filter method and has been widely used in feature extraction stage in MI-based BCI. Assume $E \in \mathbb{R}^{C \times T}$ is the multi-channel EEG, where C is the number of channels and T is the sample point. CSP obtains a projection matrix $W \in \mathbb{R}^{C \times C}$ by solving an eigenvector decomposition problem, and each column of W stands for a spatial pattern. When multichannel EEG E projects on a spatial pattern $w \in \mathbb{R}^{C \times 1}$,

$$z = w^T E \tag{2}$$

a new sequential signal $z \in \mathbb{R}^{1 \times T}$ will be obtained. Its variance character reflects the mean power information of one class. Theoretically, only after projecting on the first and last m spatial filters, variance difference of the two classes could be maximized. The variances from each projected signals z will be used as the features of one piece of multi-channel EEG. Feature computation step is explained as follows:

$$f_P = \log\left(\frac{var(z_P)}{\sum_{i=1}^{2m} var(z_i)}\right) \tag{3}$$

where $p = 1 \ldots 2m$ is the feature number. In this paper, the value of m was set to be 1 (i.e. only the first and last spatial filter in the matrix w were used for signal projection).

Figure 4 presented a top view for the power distribution of the first and last spatial filter over the scalp between the original EEG and enhanced EEG, aiming at comparing the influence to spatial filters before and after using SD-MEMD. It can be observed that the energy difference between left and right hemisphere from the enhanced EEG was larger than that of the original EEG, further proved that SD-MEMD can focus on those task-related components, so the ERD/ERS phenomenon could be more discriminant.

Support Vector Machine (SVM) with linear kernel was adopted as the classification algorithm in this experiment. For each person, there were totally 60 trials of multi-channel EEG. Forty-two trials contributed to the training set and eighteen trials left were arranged to be the test set (ratio of train and test was 0.7). Each classification result was averaged after 10-fold classification.

From Table I it can be observed that after processing with SD-MEMD, mean accuracy of thirteen out of fourteen subjects were promoted. Only Sub.8 didn't reflect any improvement. As we analyzed in section IV, Sub.8 was a typical subject who can not make a good control towards motor imagery tasks. Energy or frequency from left and right hemisphere didn't show any discriminant information. So it may be deduced that there wasn't much significant ERS/ERD phenomenon could be captured, leading some randomness existed in the classification results of this kind of subjects. Since MEMD decomposed the multi-channel EEG into sub-signals containing different frequency bands, and subject dependent selection could filter out useless components according to their individual differences, signal quality was improved and mean accuracy of these fourteen subjects was enhanced by 5.76%.

VI. Conclusion

This paper proposed a subject dependent feature extraction method for motor imagery based BCI using MEMD, so called SD-MEMD. It origins from the fact that EEG has a significant character of huge individual differences. In our study, we first utilized Multivariate Empirical Mode Decomposition (MEMD) to process the multi-channel EEG, and got a set of IMFs. Alignment could be observed in the IMF level between different channels, so the efficiency of MEMD to overcome the "unique" problem could be proved and the work of choosing IMF subset could push forward. Then, we discovered that each individual has their distinguished IMF subset to discriminate ERS/ERD, so SD-MEMD was applied for each person. At last, CSP and SVM were used to obtain features and classify the two classes of EEG. Mean accuracy was increased by 5.76% after using SD-MEMD, indicating that this method could improve the quality of MI-based EEG, and is advantageous for BCI study.

Acknowledgment

This work was supported by Shanghai Science and Technology Innovation Action Plan Project (Grant No. 16511100900), the State Key Program of National Natural Science Foundation of China (Grant No.61332008), and Shanghai NSF research project (Grant No. 16JC1401300).

References

[1] Yi, Weibo, Qiu Shuang, Qi Hong zhi, Zhang Lixin, Wan Baikun, Ming Dong. "EEG feature comparison and classification of simple and compound limb motor imagery." Journal of neuroengineering and rehabilitation vol.10(1) 2013, pp. 106-118.

[2] Kevric J, Subasi A. "Comparison of signal decomposition methods in classification of EEG signals for motor-imagery BCI system". Biomedical Signal Processing and Control, 2017, vol.31, pp. 398-406.

[3] Ramoser, Herbert, Johannes Muller-Gerking, and Gert Pfurtscheller. "Optimal spatial filtering of single trial EEG during imagined hand movement." IEEE transactions on rehabilitation engineering vol.8(4) 2000, pp. 441-446.

[4] Ang K K, Chin Z Y, Zhang H, Guan C. "Filter bank common spatial pattern (FBCSP) in brain-computer interface". Neural Networks, 2008. IJCNN 2008. (IEEE World Congress on Computational Intelligence). IEEE International Joint Conference on. IEEE, 2008, pp.2390-2397.

[5] Norden E. Huang, et al. "The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis." Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences. Vol. 454. No. 1971. The Royal Society, 1998, pp. 903-995.

[6] Zhang, De-xiang, Xiao-pei Wu, and Xiao-jing Guo. "The EEG signal preprocessing based on empirical mode decomposition." Bioinformatics and Biomedical Engineering, 2008. ICBBE 2008. The 2nd International Conference on. IEEE, 2008.

[7] He W, Wei P, Wang L, Zou Y. "A novel EMD-based common spatial pattern for motor imagery brain-computer interface". Biomedical and Health Informatics (BHI), 2012 IEEE-EMBS International Conference on. IEEE, 2012, pp. 216-219.

[8] Rehman, Naveed, and Danilo P. Mandic. "Multivariate empirical mode decomposition." Proceedings of The Royal Society of London A: Mathematical, Physical and Engineering Sciences. The Royal Society, 2009.

[9] Park, C., Looney, D., ur Rehman, N., Ahrabian, A., & Mandic, D. P. "Classification of motor imagery BCI using multivariate empirical mode decomposition." IEEE Transactions on neural systems and rehabilitation engineering vol.21(1) 2013, pp. 10-22.

[10] Gaur P, Pachori R B, Wang H, Prasad G. "A multivariate empirical mode decomposition based filtering for subject independent BCI". Signals and Systems Conference (ISSC), 2016 27th Irish. IEEE, 2016, pp. 1-7.